Program - 5

Data Preprocessing - Handling missing values, removing duplicates, applying normalization, and detecting outliers.

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Aim

To perform dataset preprocessing by handling missing values, removing duplicates, applying normalization, and detecting outliers.

Algorithm

1. Start
2. Import Pandas and NumPy libraries.
3. Create the student dataset and convert it into a DataFrame.
4. Display the original dataset and identify missing values.
5. Replace missing Marks and Attendance with their respective mean values.
6. Identify and remove duplicate records.
7. Apply Min-Max Normalization to Marks and Attendance.
8. Calculate Q1, Q3, and IQR for Marks.
9. Determine the lower and upper limits for outlier detection.
10. Identify and remove outlier records from Marks.
11. Display the final preprocessed dataset.
12. Save the cleaned dataset as preprocessed_students.csv.
13. Stop.

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```
import pandas as pd
import numpy as np

data = {
    'Student_ID': [101, 102, 103, 104, 105, 106, 106],
    'Name': ['Arun', 'Bala', 'Cathy', 'Divya', 'Ezhil', 'Fathima', 'Fathima'],
    'Marks': [85, 72, np.nan, 65, 88, 95, 95],
    'Attendance': [90, 80, 85, np.nan, 92, 95, 95]
}
df = pd.DataFrame(data)
print("Original Dataset:")
print(df)
print("\nMissing Values:")
print(df.isnull().sum())
# Replace missing Marks with mean Marks
df['Marks'] = df['Marks'].fillna(df['Marks'].mean())
# Replace missing Attendance with mean Attendance
df['Attendance'] = df['Attendance'].fillna( df['Attendance'].mean())

print("\nAfter Handling Missing Values:")
print(df)
print("\nNumber of Duplicate Records:")
print(df.duplicated().sum())
# Remove duplicate rows
df = df.drop_duplicates()

print("\nAfter Removing Duplicates:")
print(df)
# Min-Max Normalization
df['Marks_Normalized'] = ( (df['Marks'] - df['Marks'].min()) /
    (df['Marks'].max() - df['Marks'].min()) )


df['Attendance_Normalized'] = ( (df['Attendance'] - df['Attendance'].min()) /
    (df['Attendance'].max() - df['Attendance'].min()) )

print("\nAfter Min-Max Normalization:")
print(df)
Q1 = df['Marks'].quantile(0.25)
Q3 = df['Marks'].quantile(0.75)

IQR = Q3 - Q1
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```
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("\nOutlier Detection for Marks:")
print("Q1 =", Q1)
print("Q3 =", Q3)
print("IQR =", IQR)
print("Lower Limit =", lower_limit)
print("Upper Limit =", upper_limit)
# Identify outliers
outliers = df[ (df['Marks'] < lower_limit) | (df['Marks'] > upper_limit) ]

print("\nDetected Outliers:")
print(outliers)
df_clean = df[ (df['Marks'] >= lower_limit) & (df['Marks'] <= upper_limit) ]

print("\nFinal Preprocessed Dataset:")
print(df_clean)
df_clean.to_csv( 'preprocessed_students.csv', index=False )
print("\nPreprocessed dataset saved successfully!")
```